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 Studierichting: Informatica
 Oefeningenreeks 7A nr. 11.5

$$f : \Omega \subset \mathbb{R}^4 \rightarrow \mathbb{R}^2 : (w, x, y, z) \mapsto \left(\cot \left(\frac{w \ln |z|}{xy^2} \right), \ln \left| \frac{x \cos y}{(z-1)^2} \right| \right)$$

$$Df(w, x, y, z) = \begin{pmatrix} \frac{\partial f_1}{\partial w} & \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial w} & \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \end{pmatrix}$$

$$u = \frac{w \ln |z|}{xy^2}$$

$$\bullet \frac{\partial f_1}{\partial w} = -\frac{1}{\sin^2(u)} \cdot \frac{\partial}{\partial w} \left(\frac{w \ln |z|}{xy^2} \right) = -\frac{\ln |z|}{\sin^2(u)xy^2}$$

$$\bullet \frac{\partial f_1}{\partial x} = -\frac{1}{\sin^2(u)} \cdot \frac{\partial}{\partial x} \left(\frac{w \ln |z|}{xy^2} \right) = \frac{w \ln |z|}{\sin^2(u)x^2y^2}$$

$$\bullet \frac{\partial f_1}{\partial y} = -\frac{1}{\sin^2(u)} \cdot \frac{\partial}{\partial y} \left(\frac{w \ln |z|}{xy^2} \right) = \frac{2w \ln |z|}{\sin^2(u)xy^3}$$

$$\bullet \frac{\partial f_1}{\partial z} = -\frac{1}{\sin^2(u)} \cdot \frac{\partial}{\partial z} \left(\frac{w \ln |z|}{xy^2} \right) = -\frac{w}{\sin^2(u)xy^2z}$$

$$f_2 = \ln |x| + \ln |\cos y| - 2 \ln |z-1|$$

$$\bullet \frac{\partial f_2}{\partial w} = 0$$

$$\bullet \frac{\partial f_2}{\partial x} = \frac{1}{x}$$

$$\bullet \frac{\partial f_2}{\partial y} = \frac{-\sin y}{\cos y} = -\tan y$$

$$\bullet \frac{\partial f_2}{\partial z} = -2 \frac{1}{z-1} = -\frac{2}{z-1}$$

$$\Rightarrow Df(w, x, y, z) = \begin{pmatrix} -\frac{\ln |z|}{\sin^2(u)xy^2} & \frac{w \ln |z|}{\sin^2(u)x^2y^2} & \frac{2w \ln |z|}{\sin^2(u)xy^3} & -\frac{w}{\sin^2(u)xy^2z} \\ 0 & \frac{1}{x} & -\tan y & -\frac{2}{z-1} \end{pmatrix}$$

References